



ARISTOTELIS BALLAS

Aristotelis Ballas completed his Ph.D. degree in Machine and Deep Learning from the Department of Informatics and Telematics, Harokopio University of Athens. He is currently working as a Postdoctorate AI & ML researcher in several HORIZON projects. He also holds a diploma in ECE from NTUA. His research interests and publications focus on robust algorithms for representation learning, with an emphasis on OOD generalization and AI in healthcare.

PyTorch

Tensorflow

CI/CD

LaTex

EDUCATION

02/26 – CURRENT
Harokopio University
of Athens, Greece

11/21 – 12/25
Harokopio University
of Athens, Greece

10/14 – 05/21
National Technical
University of Athens,
Greece

09/11 – 06/14
2nd High School
of Glyka Nera, Greece

POSTDOC IN MACHINE & DEEP LEARNING

Title: *Learning Robust Representations for OOD Generalization via Loss Landscape Analysis*

Supervisor: *Prof. Christos Diou*

Research Interests: *OOD Generalization, Representation Learning, Biosignal Classification, AI in Healthcare*

PHD IN MACHINE & DEEP LEARNING

Dissertation: *Representation Learning Algorithms for Out-of-Distribution Generalization*

Supervisor: *Prof. Christos Diou*

DIPLOMA IN ELECTRICAL AND COMPUTER ENGINEERING

Integrated M.Sc degree in ECE (EQF Level 7)

Diploma Thesis: *Ωto_abR: A WebApp for the Visualization and Analysis of Click-Evoked Auditory Brainstem Responses*

Diploma Grade: 7.25/10.0

SCHOOL-LEAVING CERTIFICATE, 'APOLYTYRION'

Overall grade: 19.3 / 20.0

WORK EXPERIENCE

06/21 – CURRENT
Harokopio University
of Athens, Greece

AI & ML RESEARCH ASSOCIATE

AI and ML researcher for the EU funded projects of REBECCA, RELEVIMUM and MELIORA.

Involved in following Work Packages:
- REBBECA: WP2 “System design and integration” and WP4 “Causal inference and prediction using RWD”
- RELEVIMUM: WP1 “Analysis of needs and technology deployment” and WP2 “Data collection and AI-supported monitoring” and WP4 “Analysis of results and recommendations”
-MELIORA: WP5 “Technical development of the MELIORA Virtual Coach Intervention and digital tools”

05/17 – 08/21
ALMA Bank, USA

SYSTEMS ENGINEER

Provided remote IT consultation for a USA located bank from offices in Athens. Was in charge of the design, configuration and development of the domain's system and network servers along with the fine-tuning and monitoring of the infrastructure.

Also developed several automation scripts in Python and R for maintaining the infrastructure.

WORK EXPERIENCE	05/16 – 04/21 ALMA Bank, USA	HELPE\$K ANALYST Provided front-line primary technical support to end-users on various technical issues and problems relating to hardware, software and peripherals.
OTHER PROJECTS	2023 Harokopio University of Athens, Greece	HIGH-PERFOMANCE COMPUTING CLUSTER SETUP Core member of the team that was tasked to setup and administer HUA's HPC cluster. Responsibilities included: Setup of SLURM workload manager, setup of Ubuntu GPU servers, configuration of all NVIDIA GPU cards and drivers, including documentation.
TEACHING EXPERIENCE	2024 - 2025 Harokopio University of Athens, Greece	DEEP LEARNING Postgraduate course (TA) - Computer Science M.Sc
	2023 - 2024 Harokopio University of Athens, Greece	ARTIFICIAL INTELLIGENCE AND ITS APPLICATIONS ON THE INTERNET OF THINGS Postgraduate course (TA) - Computer Science M.Sc
	2022 - 2025 Harokopio University of Athens, Greece	DATA MANAGEMENT II Postgraduate course - Applied Informatics M.Sc
ACADEMIC ACTIVITIES	TUTORIALS	TUTORIAL ON DOMAIN GENERALIZATION Presented in IEEE BDS 2023, this short tutorial was designed to introduce anyone interested to the problem of Domain Generalization and assist whoever wants to get into the field, navigate the literature. Website: https://bds-dgtutorial.github.io/
	JOURNAL REVIEWER	IEEE TRANSACTIONS ON MULTIMEDIA INTERNATIONAL JOURNAL OF COMPUTER VISION PATTERN RECOGNITION NPJ DIGITAL MEDICINE NEURAL NETWORKS SIGNAL, IMAGE AND VIDEO PROCESSING
	CONFERENCE REVIEWER	IEEE ENGINEERING IN MEDICINE AND BIOLOGY ACM INTERNATIONAL CONFERENCE ON MULTIMEDIA RETRIEVAL

SKILLS + INTERESTS

Deep learning	Libraries	IT Skills	Soft Skills	Interests
PyTorch Tensorflow Keras TinyML Pytorch-Mobile	Numpy Scipy Scikit-Learn Pandas OpenCV Matplotlib	Python, R, Matlab Docker, Jenkins, Git MongoDB, SQL, InfluxDB Windows, Linux SLURM LaTeX, Wireshark, VMware	Multidisciplinary thinking Passion for learning Teamwork Able to work independently Attention to detail Project management	Cliff Diving Cinema enthusiast Tutoring Reading Dancing

LANGUAGES

Greek	English
Native	Bilingual – C2